UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576 - Empirical Process Theory and Weak Convergence Fall 2025
Final Assignment due 12/16/2025

Exercise 1 (Connectivity Threshold for Random Graphs). Let $G \sim G(n, p_n)$. In this exercise, we will pinpoint the connectivity threshold.

(i) Suppose that $p_n \geq (1+\epsilon)\frac{\ln n}{n}$. Let A_m denote the event that there exists m nodes that are disconnected from the rest of the graph. Prove that

$$\mathbb{P}[A_m] \leq \binom{n}{m}(1-p_n)^{m(n-m)}.$$

Union bound over A_m, $m = 1, \ldots, n/2$ to conclude that G is connected (whp). You may use the fact $\binom{n}{m} \leq (en/m)^m$.

(ii) Suppose that $p_n \leq (1-\epsilon)\frac{\ln n}{n}$. Use the second moment method to prove that G contains an isolated vertex (whp).

Namely, the connectivity threshold is at $\frac{\ln n}{n}$.

Exercise 2 (Bin Packing). Let $X_1, \ldots, X_n$ be iid random variables taking values in $[0, 1]$ and $A_n = f(X_1, \ldots, X_n)$ be the minimum number of bins of size 1 that can fit all the X_i. Namely, A_n is the minimum number of bins such that the sum of X_i in any given bin is always at most 1.

(a) Show that $\mathbb{E}[A_n] \geq n \cdot \mathbb{E}[X_1]$.

(b) Show that $\mathrm{Var}(A_n) \leq n/2$.

(c) If $\mathbb{E}[X_1] = t/\sqrt{n}$ for $t > 1$, prove that $\mathbb{P}[A_n > \sqrt{n}] \geq 1 - 2e^{-2(t-1)^2}$.

Exercise 3 (ℓ_0 Minimization). Suppose $X \in \mathbb{R}^{n\times p}$ is $(\delta, 2s)$-RIP for $\delta \in (0, \frac{1}{3})$. We established in the lecture that every s-sparse vector β is the unique optimal solution to $\min \|z\|_1$ subject to $Xz = X\beta$. Establish that the same is also true for the objective $\|z\|_0$. Namely, under the same assumptions on X,

$$\min_{z\in\mathbb{R}^p} \|z\|_0$$
$$\text{Subject to } Xz = X\beta.$$

admits a unique optimal solution, which is β.

Exercise 4 (Coherence and Deterministic RIP Matrices). In this exercise, we provide an explicit construction of a matrix satisfying the RIP, without resorting to the randomness involved in our earlier example with iid Gaussian entries.

(a) Given a matrix $X \in \mathbb{R}^{n\times p}$ with columns $X_1, \ldots, X_p$ having unit norm, $\|X_i\|_2 = 1$, define its *coherence* as $\mu(X) := \max_{1\le i<j\le p} |\langle X_i, X_j\rangle|$. Prove that if $\mu(X) \le \delta/s$, then X is (δ, s)-RIP.

Now, fix a prime number z and a positive integer r. Consider an $z^2 \times z^{r+1}$ matrix Φ constructed as follows. Each row of Φ is indexed by a pair (x, y) of integers $0 \le x, y \le z-1$.[1] Each column is indexed by a polynomial of the form $f(t) = a_0 + a_1 t + a_2 t^2 + \cdots + a_r t^r$ where $a_j, 0 \le j \le r$ take values in $\{0, 1, \ldots, z-1\}$.[2] For every entry of the matrix indexed by row (x, y) and column f we put value $1/\sqrt{z}$ if $y \equiv f(x) \pmod z$ and value 0 otherwise.[3]

(b) Establish that the $\|\cdot\|_2$ norm of each column of Φ is 1.

(c) Use (a) to conclude that for every δ, Φ is (δ, s)-RIP with $s = z\delta/r$.

Note. This construction is due to DeVore [DeV07].

Exercise 5 (ℓ_1-Recovery). Given $A \in \mathbb{R}^{n\times p}$ and $w \in \mathbb{R}^n$, consider

$$\begin{aligned} &\min \|z\|_1 \\ &\text{subject to : } Az = w. \end{aligned} \tag{1}$$

Suppose $s \le p$, $n \ge Cs\log(p/s)$ for some $C > 0$, and A consists of iid $\mathcal{N}(0,1)$ entries. Let $w = Az^* + \frac{1}{\sqrt{s}} A_S e_S$, where $z^* \in \mathbb{R}^p$ is any s-sparse vector, $S \subset [p]$ is any subset with $|S| = s$, and $e_S = (1, 1, \ldots, 1) \in \mathbb{R}^s$ is a vector of all ones. As usual, A_S denotes here the $n \times s$ matrix induced by columns in S.

Let $\bar{z}$ the any solution to the optimization problem (1). Prove that for a sufficiently large constant C, it holds that $\|\bar{z} - z^*\|_2 \le 1$ with high probability over the entries of A as $p \to \infty$.

Note: The statement holds for any choice of z^* and S subject to the restrictions above. In particular, S is not necessarily the support of z^*.

[1] So, there are indeed z^2 rows.

[2] So altogether we indeed have z^{r+1} such polynomials.

[3] We say $y \equiv f(x) \pmod z$ if $y - f(x)$ is divisible by z.

Exercise 6 (Gaussian Smart Path Interpolation). Consider two independent random vectors $X \sim \mathcal{N}(0, \Sigma_1)$ and $Y \sim \mathcal{N}(0, \Sigma_2)$. Define

$$Z(t) := \sqrt{t}X + \sqrt{1-t}Y, \quad t \in [0,1].$$

Show that for any twice-differentiable $f : \mathbb{R}^n \to \mathbb{R}$, we have

$$\frac{d}{dt}\mathbb{E}f\big(Z(t)\big) = \frac{1}{2}\left\langle \Sigma_1 - \Sigma_2 \,,\, \mathbb{E}\big[\nabla^2 f\big(Z(t)\big)\big] \right\rangle,$$

where $\langle M, N \rangle = \sum_{i,j} M_{ij}N_{ij}$ is the matrix inner product and $\nabla^2 f \in \mathbb{R}^{n\times n}$ is the Hessian of f, i.e., $(\nabla^2 f)_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$. (Assume all expectations are finite.)

Hint: You can use multivariate Gaussian integration by parts. Namely, for any such $f : \mathbb{R}^n \to \mathbb{R}$ and $X \sim \mathcal{N}(0, \Sigma)$, $\mathbb{E}[Xf(X)] = \Sigma \cdot \mathbb{E}\nabla f(X)$:

$$\mathbb{E}[X_i f(X)] = \sum_{j=1}^{n} \Sigma_{ij} \mathbb{E}\frac{\partial f}{\partial x_j}(X), \quad i = 1, \ldots, n.$$

Exercise 7 (Gaussian Width of Image Sets). Let $A \in \mathbb{R}^{m\times n}$ be a matrix, T be an arbitrary subset of $\mathbb{R}^n$ and AT be the image set $\{Ax : x \in T\}$. Establish that

$$w(AT) \le \|A\| \cdot w(T),$$

where w is the Gaussian width, $w(T) = \mathbb{E}_{g\sim\mathcal{N}(0,I_n)}\left[\sup_{x\in T}\langle x, g\rangle\right]$.

Exercise 8 (Chevet-like Inequality). Let $A \in \mathbb{R}^{m\times n}$ be a matrix with iid $\mathcal{N}(0,1)$ entries and T be a bounded subset of $\mathbb{R}^n$ which contains the origin. Show that

$$\mathbb{E}\left[\mathrm{rad}\left(\frac{1}{\sqrt{n}}AT\right)\right] \le \sqrt{m/n} \cdot \mathrm{rad}(T) + Cw(T)/\sqrt{n},$$

where $\mathrm{rad}(T) = \sup_{x\in T}\|x\|_2$ and AT is the *image* of T under A (as in Exercise 7). *Hint:* Apply Sudakov-Fernique to compare with a suitable process.

Exercise 9 ($\ell_p \to \ell_q$ **Operator Norm).** Let $A \in \mathbb{R}^{m\times n}$ with iid $\mathcal{N}(0,1)$ entries and recall that $\|x\|_p := (\sum_i |x_i|^p)^{1/p}$. The $\ell_p \to \ell_q$ norm of A is defined by

$$\|A\|_{p\to q} := \sup_{x:\|x\|_p=1} \|Ax\|_q.$$

Define p' such that $1/p + 1/p' = 1$ and similarly for q'.

(a) Show, by Holder's inequality, that for a general matrix A,

$$\|A\|_{p\to q} = \sup_{\substack{x:\|x\|_p=1 \\ y:\|y\|_{q'}=1}} \langle Ax, y\rangle.$$

(b) Suppose that $1 < p \le 2$ and $2 \le q < \infty$. Show that for a constant $C := C_{p,q} > 0$ depending only on p and q, we have:

$$\mathbb{E}\|A\|_{p\to q} \le C_{p,q}\left(n^{1/p'} + m^{1/q}\right).$$

Hint: Recall that if B_p is the unit ball in ℓ_p norm, $w(B_p) \le C\sqrt{p'}n^{1/p'}$, as shown in Homework 5.

Exercise 10 (Bonus Problem- Sudakov-Fernique Inequality). Suppose $(X_t)_{t\in T}$ and $(Y_t)_{t\in T}$ are two zero-mean Gaussian processes and assume $\mathbb{E}(X_t - X_s)^2 \le \mathbb{E}(Y_t - Y_s)^2$ for every $t, s \in T$. We will prove Sudakov-Fernique inequality: $\mathbb{E}\sup_t X_t \le \mathbb{E}\sup_t Y_t$. Let $T = \{1, \dots, n\}$. Fix a $\beta > 0$ and consider *soft-max*

$$f_\beta(x) = \frac{1}{\beta}\log\sum_{i=1}^n e^{\beta x_i}.$$

(a) Differentiate f_β to establish that

$$\frac{\partial f_\beta}{\partial x_i} = \frac{e^{\beta x_i}}{\sum_k e^{\beta x_k}} := p_i(x) \text{ and } \frac{\partial^2 f_\beta}{\partial x_i \partial x_j} = \beta\big(\mathbf{1}\{i=j\}p_i(x) - p_i(x)p_j(x)\big).$$

(b) Define $Z(t) := \sqrt{t}X + \sqrt{1-t}Y$. Use Gaussian interpolation (Exercise 6) to argue that $\frac{d}{dt}\mathbb{E}f\big(Z(t)\big) \le 0$.

(c) Take $\beta \to \infty$ to conclude the proof.

References

[DeV07] Ronald A DeVore, *Deterministic constructions of compressed sensing matrices*, Journal of complexity **23** (2007), no. 4-6, 918–925.